

Subnetting Practice

Network Requirements · CIDR Notation · IPv4

1

28.0.0.0/8 → 60 Networks

Class A · Original prefix /8

1

FIND THE REQUIRED SUBNET BITS ($2^N \geq \text{NETWORKS NEEDED}$)

Need ≥ 60 networks

Formula: $2^n \geq 60$

Try $n = 5$: $2^5 = 32$ ✗ not enough

Try $n = 6$: $2^6 = 64$ ✓ sufficient

∴ Subnet bits $n = 6$

2

CALCULATE THE NEW PREFIX LENGTH (ORIGINAL + N)

Original prefix = /8 (8 network bits)

New prefix = original + n = $8 + 6 = 14$

∴ New CIDR notation = /14

3

DERIVE THE SUBNET MASK (BINARY → DECIMAL)

/14 binary: 11111111 . 11111100 . 00000000 . 00000000



2nd octet: $128 + 64 + 32 + 16 + 8 + 4 = 252$

∴ Subnet mask = 255 . 252 . 0 . 0

4

BLOCK SIZE, SUBNET COUNT, AND USABLE HOSTS

Block size = $2^{(32-14)} = 2^{18} = 262,144$ (spans +4 in 2nd octet)

Borrowed bits = new prefix - orig = $14 - 8 = 6$

Number of subnets = $2^6 = 64$

Usable hosts/net = $2^{18} - 2 = 262,142$

STEP 5 · FIRST 5 AND LAST OF 64 SUBNETS (BLOCK = +4 IN 2ND OCTET)

#	NETWORK ADDRESS	FIRST HOST	LAST HOST	BROADCAST
1	28.0.0.0/14	28.0.0.1	28.3.255.254	28.3.255.255

#	NETWORK ADDRESS	FIRST HOST	LAST HOST	BROADCAST
2	28.4.0.0/14	28.4.0.1	28.7.255.254	28.7.255.255
3	28.8.0.0/14	28.8.0.1	28.11.255.254	28.11.255.255
4	28.12.0.0/14	28.12.0.1	28.15.255.254	28.15.255.255
5	28.16.0.0/14	28.16.0.1	28.19.255.254	28.19.255.255
...	... /14	...	...	...
64	28.252.0.0/14	28.252.0.1	28.255.255.254	28.255.255.255

Full pattern: 28.X.0.0/14 where X = 0, 4, 8, 12, ... , 252

2

172.16.0.0/16 → 45 Networks

Class B · Original prefix /16

1 FIND THE REQUIRED SUBNET BITS ($2^n \geq \text{NETWORKS NEEDED}$)

Need ≥ 45 networks

Formula: $2^n \geq 45$

Try $n = 5$: $2^5 = 32$ × not enough

Try $n = 6$: $2^6 = 64$ ✓ sufficient

∴ Subnet bits $n = 6$

2 CALCULATE THE NEW PREFIX LENGTH (ORIGINAL + N)

Original prefix = /16 (16 network bits)

New prefix = original + n = 16 + 6 = 22

∴ New CIDR notation = /22

3 DERIVE THE SUBNET MASK (BINARY → DECIMAL)

/22 binary: 11111111 . 11111111 . 11111100 . 00000000



3rd octet: $128 + 64 + 32 + 16 + 8 + 4 = 252$

∴ Subnet mask = 255 . 255 . 252 . 0

4 BLOCK SIZE, SUBNET COUNT, AND USABLE HOSTS

Block size = $2^{(32-22)} = 2^{10} = 1,024$ (spans +4 in 3rd octet)

Borrowed bits = new prefix - orig = 22 - 16 = 6

Number of subnets = $2^6 = 64$

Usable hosts/net = $2^{10} - 2 = 1,022$

STEP 5 · FIRST 5 AND LAST OF 64 SUBNETS (BLOCK = +4 IN 3RD OCTET)

#	NETWORK ADDRESS	FIRST HOST	LAST HOST	BROADCAST
1	172.16.0.0/22	172.16.0.1	172.16.3.254	172.16.3.255
2	172.16.4.0/22	172.16.4.1	172.16.7.254	172.16.7.255
3	172.16.8.0/22	172.16.8.1	172.16.11.254	172.16.11.255
4	172.16.12.0/22	172.16.12.1	172.16.15.254	172.16.15.255
5	172.16.16.0/22	172.16.16.1	172.16.19.254	172.16.19.255
...	... /22	...	...	...
64	172.16.252.0/22	172.16.252.1	172.16.255.254	172.16.255.255

Full pattern: 172.16.X.0/22 where X = 0, 4, 8, 12, ... , 252

3

210.40.20.0/24 → 12 Networks

Class C · Original prefix /24

1 FIND THE REQUIRED SUBNET BITS ($2^n \geq \text{NETWORKS NEEDED}$)

Need ≥ 12 networks

Formula: $2^n \geq 12$

Try $n = 3$: $2^3 = 8$ ✗ not enough

Try $n = 4$: $2^4 = 16$ ✓ sufficient

∴ Subnet bits $n = 4$

2 CALCULATE THE NEW PREFIX LENGTH (ORIGINAL + N)

Original prefix = /24 (24 network bits)

New prefix = original + n = 24 + 4 = 28

∴ New CIDR notation = /28

3 DERIVE THE SUBNET MASK (BINARY → DECIMAL)

/28 binary: 11111111 . 11111111 . 11111111 . 11110000

255 255 255 ?

4th octet: $128 + 64 + 32 + 16 = 240$

∴ Subnet mask = 255 . 255 . 255 . 240

4 BLOCK SIZE, SUBNET COUNT, AND USABLE HOSTS

Block size = $2^{(32-28)} = 2^4 = 16$

Borrowed bits = new prefix - orig = 28 - 24 = 4

Number of subnets = $2^4 = 16$

Usable hosts/net = $2^4 - 2 = 14$

STEP 5 · ALL 16 SUBNETS (BLOCK SIZE = 16)				
#	NETWORK ADDRESS	FIRST HOST	LAST HOST	BROADCAST
1	210.40.20.0/28	210.40.20.1	210.40.20.14	210.40.20.15
2	210.40.20.16/28	210.40.20.17	210.40.20.30	210.40.20.31
3	210.40.20.32/28	210.40.20.33	210.40.20.46	210.40.20.47
4	210.40.20.48/28	210.40.20.49	210.40.20.62	210.40.20.63
5	210.40.20.64/28	210.40.20.65	210.40.20.78	210.40.20.79
6	210.40.20.80/28	210.40.20.81	210.40.20.94	210.40.20.95
7	210.40.20.96/28	210.40.20.97	210.40.20.110	210.40.20.111
8	210.40.20.112/28	210.40.20.113	210.40.20.126	210.40.20.127
9	210.40.20.128/28	210.40.20.129	210.40.20.142	210.40.20.143
10	210.40.20.144/28	210.40.20.145	210.40.20.158	210.40.20.159
11	210.40.20.160/28	210.40.20.161	210.40.20.174	210.40.20.175
12	210.40.20.176/28	210.40.20.177	210.40.20.190	210.40.20.191
13	210.40.20.192/28	210.40.20.193	210.40.20.206	210.40.20.207
14	210.40.20.208/28	210.40.20.209	210.40.20.222	210.40.20.223
15	210.40.20.224/28	210.40.20.225	210.40.20.238	210.40.20.239
16	210.40.20.240/28	210.40.20.241	210.40.20.254	210.40.20.255

■■ FINAL ANSWERS · ALL THREE PROBLEMS SOLVED ■■								
#	NETWORK	NEED	N BITS	PREFIX	SUBNET MASK	BLOCK	SUBNETS	USABLE HOSTS
1	28.0.0.0/8	60	6	/14	255.252.0.0	262,144	64	262,142
2	172.16.0.0/16	45	6	/22	255.255.252.0	1,024	64	1,022
3	210.40.20.0/24	12	4	/28	255.255.255.240	16	16	14